Directionally Aligned Federated Aggregation: Improving Stability and Signal-to-Noise Ratio under Heterogeneity

Yashuo Jia

March 2026

Abstract

Federated learning (FL) enables collaborative model training across decentralized clients without sharing raw data. However, under heterogeneous (non-IID) data distributions, client updates often exhibit severe directional disagreement. Classical aggregation methods, such as Federated Averaging (FedAvg), treat client updates isotropically, ignoring the directional reliability of individual gradients. This leads to destructive interference, poor signal-to-noise ratios (SNR), and unstable convergence.

In this work, we analyze federated optimization through the lens of directional alignment and introduce the **Directional Signal-to-Noise Ratio (DSNR)** as a principled metric for evaluating aggregated updates. Based on this perspective, we propose **Directionally Aligned Federated Aggregation (DAFA)**, a novel aggregation framework that assigns adaptive weights to client updates based on their directional agreement with a proxy global descent direction.

We provide a rigorous theoretical framework demonstrating that DAFA: (1) provably improves the expected **DSNR lower bound** compared to uniform averaging by formulating a tractable surrogate objective; (2) introduces a strictly bounded aggregation bias with explicit dependence on the softmax temperature; and (3) achieves standard non-convex convergence guarantees, reaching $O(1/\sqrt{T})$ convergence under diminishing step sizes, and provably reduces the heterogeneity constant under mild alignment assumptions. Empirical evaluations on standard federated benchmarks (CIFAR-10, FEMNIST, Shakespeare) confirm our theoretical findings, demonstrating that DAFA significantly improves training stability and convergence speed under severe data heterogeneity while incurring zero additional communication overhead.

1 Introduction

Federated learning (FL) has emerged as a prevailing paradigm for distributed machine learning, allowing a central server to train a global model by aggregating locally computed updates from a federation of clients [1]. While highly

successful in protecting data privacy, FL faces a fundamental challenge in practical deployments: **statistical heterogeneity**.

When local data distributions are highly non-IID, local optimization steps diverge significantly. The de facto standard aggregation algorithm, **Federated Averaging (FedAvg)**, aggregates these divergent updates using fixed weights proportional to client dataset sizes. While computationally simple, FedAvg implicitly assumes that local updates are approximately aligned. Under strong heterogeneity, this assumption breaks down. The aggregation of disparate updates leads to severe **signal cancellation and noise amplification**, manifesting as slow convergence and training instability [2, 3].

Recent works have proposed various mechanisms to mitigate client drift, including proximal regularization (FedProx [2]), control variates (SCAFFOLD [3]), and normalized aggregation (FedNova [4]. However, these approaches primarily operate in the **magnitude domain** (e.g., step normalization or proximal penalties) or rely on **auxiliary state variables** maintained at clients. They do not explicitly model the **directional reliability** of client updates.

In this paper, we argue that the key bottleneck in heterogeneous federated optimization is **directional inconsistency**. We observe that the effectiveness of an aggregated update depends on two competing factors: the *signal* (its alignment with the true global descent direction) and the *noise* (the variance and drift induced by heterogeneous clients).

To formalize this intuition, we introduce the **Directional Signal-to-Noise Ratio (DSNR)**. Directly maximizing the fractional DSNR is intractable in the federated setting. Instead, we derive a rigorous **surrogate lower bound** that transforms DSNR maximization into a directional alignment problem. This leads to our proposed algorithm, **Directionally Aligned Federated Aggregation (DAFA)**, which dynamically applies a softmax weighting over client updates based on their cosine similarity with a server-side proxy direction.

Our main contributions are summarized as follows:

1. **Directional Perspective.** We introduce DSNR to quantify the directional quality of aggregated federated updates, and derive a provable **lower-bound surrogate** that is tractable to optimize.

2. **Algorithm Design.** We propose DAFA, a communication-efficient aggregation rule that dynamically downweights discordant updates to suppress orthogonal gradient noise. DAFA is purely server-side, introduces no extra communication, and recovers FedAvg when the softmax temperature goes to zero.

3. **Theoretical Guarantees.** We provide rigorous proofs showing that DAFA improves a DSNR lower bound, controls aggregation bias with explicit dependence on the softmax temperature, and achieves an $\mathcal{O}(1/\sqrt{T})$ non-convex convergence rate with a strictly smaller heterogeneity constant than FedAvg.

4. **Empirical Validation.** We present comprehensive experiments on CIFAR-10, FEMNIST, and Shakespeare under strong non-IID settings. Results show that DAFA substantially improves convergence speed, final accuracy, and empirical DSNR, and remains robust to hyperparameter choices.

—

2 Related Work

2.1 Federated Optimization under Heterogeneity

Since the introduction of FedAvg [1], tackling data heterogeneity has been a focal point of FL research. **FedProx** [2] introduces a proximal term to regularize local updates towards the global model, effectively adding a quadratic penalty on the distance between local and global parameters. **SCAFFOLD** [3] uses control variates maintained at both clients and server to correct client drift, at the cost of storing and communicating additional control vectors. **FedNova** [4] normalizes local updates to eliminate objective inconsistency caused by varying local steps and learning rates.

These methods primarily operate on **update magnitude** (e.g., scaling local steps, constraining parameter distance) or on **auxiliary control states**. In contrast, DAFA focuses on the **direction** of updates and is implemented entirely on the server without modifying client-side optimization.

2.2 Gradient Conflict Resolution in Multi-Task Learning

Gradient conflict is a well-known issue in multi-task learning (MTL). **PCGrad** [5] projects conflicting task gradients onto the normal plane of others to avoid negative transfer, while **CAGrad** [6] optimizes a worst-case objective in a local neighborhood. These methods perform fine-grained gradient surgery across tasks.

Directly applying such gradient surgery to cross-device FL is challenging due to the **large number of clients**, **communication constraints**, and the dynamic client participation pattern. DAFA draws inspiration from these methods' focus on directional conflict but employs a **lightweight softmax weighting** that scales to thousands or millions of clients.

2.3 Directional / Projection-Based Aggregation in FL

Several recent works implicitly or explicitly use directional cues in FL. For example, certain variants of FedProx can be interpreted as projecting client updates onto a feasible set defined by proximity constraints [2]. Methods that normalize gradient magnitudes (e.g., FedNova [4]) partially reduce directional conflict, but do not explicitly optimize for alignment.

More directly, some works adjust aggregation weights based on similarity of client updates (e.g., cosine similarity) or on alignment of model outputs across classes. These methods typically require additional per-client state or communication.

DAFA's key distinguishing features are:

- **Explicit DSNR-based motivation**: it directly links directional alignment to a lower bound of DSNR.
- **Purely server-side implementation**: no client-side changes or extra states.

- **Theoretical analysis**: it provides explicit bias bounds and an improved heterogeneity constant.

3 Problem Setup and Motivation

We consider the standard federated learning objective over N clients:

$$\min_{x \in R^d} F(x) = \sum_{i=1}^{N} p_i F_i(x),$$

where $F_i(x)$ is the local expected loss on client i and $p_i \geq 0$, $\sum_{i=1}^{N} p_i = 1$ is proportional to its dataset size.

At communication round t, the server broadcasts the global model x_t to a subset of clients $\mathcal{S}_t$. Each selected client performs K local SGD steps, obtaining $x_{i,K}^t$ and the update

$$\Delta_i^t = x_{i,K}^t - x_t.$$

The server aggregates these updates:

$$\Delta_{agg}^t = \sum_{i \in \mathcal{S}_t} w_i^t \Delta_i^t, \quad x_{t+1} = x_t + \eta \Delta_{agg}^t.$$

FedAvg corresponds to $w_i^t = p_i$ (or proportional to local dataset size in $\mathcal{S}_t$).

Directional Dilemma. Under non-IID data, multi-step local training introduces significant **client drift**: each Δ_i^t may point toward a local optimum instead of the global one. When directions among clients diverge, uniform averaging can **cancel signal** and **amplify noise**, resulting in a small, misaligned aggregated step, and noisy convergence.

Our goal is to **adaptively reweight** client updates based on their *directional reliability* while preserving communication efficiency and theoretical tractability.

4 Directional Signal-to-Noise Ratio (DSNR)

Let the ideal global update at round t be

$$\Delta_*^t = -\eta_{global} \nabla F(x_t), \quad v_{true}^t = \frac{\Delta_*^t}{\|\Delta_*^t\|}.$$

We decompose each client update as

$$\Delta_i^t = \Delta_*^t + d_i^t + \epsilon_i^t,$$

where d_i^t denotes deterministic **client drift** due to data heterogeneity, and ϵ_i^t denotes zero-mean stochastic noise from local SGD.

For an aggregated update $\Delta_{agg}^t = \sum_i w_i^t \Delta_i^t$, we define its **Directional Signal-to-Noise Ratio (DSNR)** as

$$DSNR(\Delta_{agg}^t) = \frac{\left(E_\epsilon[\langle \Delta_{agg}^t, v_{true}^t \rangle]\right)^2}{E_\epsilon[\|\Delta_{agg}^t - \Delta_*^t\|^2]}.$$

- **Numerator**: squared projection of the expected aggregated update along the true descent direction the usable signal.
- **Denominator**: mean squared deviation from the ideal update — combining drift and stochastic noise.

A higher DSNR indicates a more reliable descent step.

Remark (Unobservability and Surrogate). In practice, v_{true}^t is unavailable, hence true DSNR cannot be computed exactly. In Section 6 we introduce a **tractable lower bound** on DSNR via alignment with a **proxy direction** v_*^t, and our theoretical guarantees and algorithm design are built around this **DSNR lower bound**, not the exact DSNR.

5 Directionally Aligned Federated Aggregation (DAFA)

5.1 Proxy Direction Estimation

Since v_{true}^t is inaccessible, we estimate a **proxy direction** v_*^t via a momentum-smoothed FedAvg update:

$$\Delta_{\text{FedAvg}}^t = \frac{1}{|\mathcal{S}_t|} \sum_{i \in \mathcal{S}_t} \Delta_i^t,$$

$$m^t = (1-\beta)\, \Delta_{\text{FedAvg}}^t + \beta\, m^{t-1}, \quad v_*^t = \frac{m^t}{\|m^t\|},$$

where $\beta \in [0, 1)$ is a momentum coefficient and $m^{-1} = 0$.

Intuitively, m^t aggregates information across rounds and stabilizes the direction estimate.

5.2 Direction-Aware Softmax Weighting

For each client $i \in \mathcal{S}_t$, we normalize its update to obtain a **direction vector**. To avoid amplifying local noise near stationary points (as dictated by Assumption 3), we introduce a magnitude threshold μ:

$$v_i^t = \begin{cases} \frac{\Delta_i^t}{\|\Delta_i^t\|}, & \text{if } \|\Delta_i^t\| \geq \mu \\ 0, & \text{otherwise.} \end{cases}$$

We then compute its **alignment score** with the proxy direction:

$$s_i^t = \langle v_i^t, v_*^t \rangle \in [-1, 1].$$

To preserve statistical efficiency and fairness across clients with varying dataset sizes, we compute the aggregation weights by scaling the softmax scores with the client's original data proportion p_i:

$$w_i^t = \frac{p_i \exp(\gamma s_i^t)}{\sum_{j \in \mathcal{S}_t} p_j \exp(\gamma s_j^t)}.$$

- $\gamma = 0$ perfectly recovers standard data-size weighted averaging (FedAvg).
- **Graceful Degradation:** For clients with very small, noisy updates ($\|\Delta_i^t\| < \mu$), their score is set to $s_i^t = 0$. Consequently, their aggregation weight effectively receives a neutral baseline multiplier ($\exp(0) = 1$), naturally down-weighting them relative to highly-aligned clients and preventing the dangerous over-amplification of pure noise.
- Larger γ enforces stronger preference for well-aligned clients...

Algorithm 1 Directionally Aligned Federated Aggregation (DAFA)

Initialize global model x_0, proxy momentum $m_{-1} = 0$
Initialize hyperparameters γ, β, η, and threshold μ
for $t = 0$ to $T - 1$ **do**
  Server samples a subset of clients $\mathcal{S}_t$ and broadcasts x_t
  for $i \in \mathcal{S}_t$ (in parallel) **do**
    Perform K local SGD steps on F_i to get $x_{i,K}^t$
    Compute local update $\Delta_i^t = x_{i,K}^t - x_t$
    Send Δ_i^t to server
  end for
  Server computes standard update: $\Delta_{\text{FedAvg}}^t = \frac{1}{\sum_{i \in \mathcal{S}_t} p_i} \sum_{i \in \mathcal{S}_t} p_i \Delta_i^t$
  Update proxy momentum: $m_t = (1 - \beta)\Delta_{\text{FedAvg}}^t + \beta m_{t-1}$
  Compute proxy direction: $v_t^* = m_t / \|m_t\|$
  for $i \in \mathcal{S}_t$ **do**
    if $\|\Delta_i^t\| \geq \mu$ **then**
      Compute $v_i^t = \Delta_i^t / \|\Delta_i^t\|$ and $s_i^t = \langle v_i^t, v_t^* \rangle$
    else
      $v_i^t = 0$ and $s_i^t = 0$ ▷ Safeguard against noise amplification
    end if
    Compute weights: $w_i^t = \dfrac{p_i \exp(\gamma s_i^t)}{\sum_{j \in \mathcal{S}_t} p_j \exp(\gamma s_j^t)}$
  end for
  Aggregate update: $\Delta_{\text{DAFA}}^t = \sum_{i \in \mathcal{S}_t} w_i^t \Delta_i^t$
  Update global model: $x_{t+1} = x_t + \eta \Delta_{\text{DAFA}}^t$
end for

6 Theoretical Analysis

We analyze DAFA under standard assumptions (smoothness, bounded variance, bounded drift) plus two additional mild assumptions specific to the directional setting.

6.1 Assumptions

We omit superscript t when clear.

1. **L-Smoothness.** F and all F_i are L-smooth:

$$\|\nabla F(x) - \nabla F(y)\| \leq L\|x - y\|.$$

2. **Independent Bounded Variance.** Stochastic noise ϵ_i is independent and zero-mean with bounded variance:

$$E[\epsilon_i] = 0, \quad E[\|\epsilon_i\|^2] \leq \sigma^2.$$

3. **Bounded Update Magnitudes.** There exist constants $0 < \mu \leq G$ such that

$$\mu \leq \|\Delta_i\| \leq G.$$

This is natural away from stationary points. Near convergence, we can enforce this via thresholding.

4. **Bounded Client Drift.** Deterministic drift satisfies

$$E[\|d_i\|^2] \leq \delta^2.$$

5. **Proxy Direction Reliability.** The proxy direction has positive expected correlation with the true direction:

$$E[\langle v_*, v_{true}\rangle] \geq \rho > 0.$$

Section 7 empirically validates this assumption.

6.2 DSNR Surrogate Lower Bound

Instead of maximizing the intractable true DSNR, DAFA optimizes a **Proxy DSNR** computed with respect to the accessible proxy direction v_*.

Theorem 6.1 (Surrogate DSNR Bound)**.** *Under Assumptions 2–4, taking the total expectation E over both stochastic noise and client sampling, the Proxy DSNR of Δ_{agg} is lower bounded by*

$$DSNR_{proxy}(\Delta_{agg}) := \frac{E[\langle \Delta_{agg}, \Delta_*\rangle]^2}{E[\|\Delta_{agg} - \Delta_*\|^2]} \geq C \left(\sum_{i \in \mathcal{S}_t} w_i \langle v_i, v_*\rangle \right)^2,$$

where $C_t = \frac{\mu^2 \|\Delta_*^t\|^2}{\delta^2 + \sigma^2} > 0$ is a bounded scalar at round t, strictly independent of the aggregation weights w_i^t.

Sketch. Let the aggregated update be $\Delta_{\text{agg}} = \sum_i w_i(\Delta_* + d_i + \epsilon_i)$. The numerator evaluates the alignment with the proxy direction. Using $\Delta_i = \|\Delta_i\| v_i$ and the magnitude lower bound (Assumption 3):

$$E[\langle \Delta_{\text{agg}}, \Delta_*\rangle]^2 = \|\Delta_*\|^2 \left(E\left[\sum_i w_i \|\Delta_i\| \langle v_i, v_*\rangle \right] \right)^2 \geq \mu^2 \|\Delta_*\|^2 \left(E\left[\sum_i w_i \langle v_i, v_*\rangle \right] \right)^2.$$

For the denominator, we account for both stochastic noise ϵ_i and deterministic client drift d_i. Since $\sum_i w_i = 1$:

$$\Delta_{\text{agg}} - \Delta_* = \sum_i w_i d_i + \sum_i w_i \epsilon_i.$$

By the independence of ϵ_i (Assumption 2) and applying Jensen's inequality to the bounded drift over client sampling (Assumption 4):

$$E\left[\|\Delta_{\text{agg}} - \Delta_*\|^2\right] = E\left[\left\|\sum_i w_i d_i\right\|^2\right] + \sum_i w_i^2 E[\|\epsilon_i\|^2].$$

Using the convexity of the squared norm and $w_i \in [0,1]$, we have $\|\sum_i w_i d_i\|^2 \leq \sum_i w_i \|d_i\|^2$. Taking the expectation yields $\sum_i w_i E[\|d_i\|^2] \leq \sum_i w_i \delta^2 = \delta^2$. Furthermore, $\sum_i w_i^2 \sigma^2 \leq (\max_i w_i) \sum_i w_i \sigma^2 \leq \sigma^2$. Thus, the denominator is strictly upper bounded by $\delta^2 + \sigma^2$. Combining the bounds yields the theorem.

6.3 Bounded Aggregation Bias

Because DAFA uses data-dependent weights, the aggregated update is generally a **biased estimator** of Δ_*. We show this bias is controlled and explicitly depends on the softmax temperature.

Let w^* denote uniform weights and $e = \Delta_{\text{DAFA}} - \Delta_*$ the aggregation error.

Theorem 6.2 (Bounded Aggregation Bias)**.** *Under Assumptions 2–4, the bias satisfies*

$$\left\|E[\Delta_{DAFA}] - \Delta_*\right\| \leq \mathcal{O}\big(\gamma G \sqrt{\delta^2 + \sigma^2}\big) + \delta.$$

Sketch. Decompose

$$\sum_i w_i(\Delta)\Delta_i - \Delta_* = \sum_i (w_i(\Delta) - w_i^*)\Delta_i + \Big(\sum_i w_i^* \Delta_i - \Delta_*\Big).$$

The second term is bounded by δ. For the first term,

$$\Big\|\sum_i (w_i(\Delta) - w_i^*)\Delta_i\Big\| \leq G\|w(\Delta) - w^*\|_1.$$

Softmax with temperature γ is γ-Lipschitz in ℓ_1 norm; perturbations in scores s_i due to d_i, ϵ_i are bounded by $\mathcal{O}(\|d_i\| + \|\epsilon_i\|)$. Taking expectation yields the bound.

This result formalizes the **bias-variance trade-off**: larger γ yields stronger alignment (better DSNR lower bound) but increases bias linearly in γ.

6.4 Non-Convex Convergence

Let $\Delta_{\text{DAFA}} = -\nabla F(x_t) + e_t$, where e_t encodes both aggregation bias and stochastic variance.

Theorem 6.3 (Convergence Rate)**.** *Under Assumptions 1–5, with a constant step size $\eta \leq 1/(4L)$, DAFA converges to a neighborhood of a stationary point:*

$$\min_{t<T} E[\|\nabla F(x_t)\|^2] \leq \mathcal{O}\left(\frac{F(x_0) - F^*}{\eta T} + \eta L(\sigma^2 + \Gamma_{DAFA})\right),$$

where $\Gamma_{DAFA} = E\left[\sum_i w_i \|d_i\|^2\right]$ is the heterogeneity constant. ***Corollary****: Using a diminishing step size $\eta_t = \mathcal{O}(1/\sqrt{t})$, DAFA achieves an exact asymptotic convergence rate of $\mathcal{O}(1/\sqrt{T})$.*

Improved Heterogeneity Constant. For FedAvg, $\Gamma_{\text{FedAvg}} = E[\sum_i p_i \|d_i\|^2]$. In DAFA, misaligned clients (which typically exhibit larger deterministic drift $\|d_i\|$) are assigned exponentially smaller weights. Under the mild assumption that client drift magnitude $\|d_i\|$ is negatively correlated with the proxy alignment score s_i, we have:

$$\Gamma_{\text{DAFA}} = E\left[\sum_i w_i \|d_i\|^2\right] < E\left[\sum_i p_i \|d_i\|^2\right] = \Gamma_{\text{FedAvg}}.$$

This demonstrates DAFA's ability to selectively suppress severe client drift.

Sketch. Standard smoothness analysis yields a descent lemma with an additional term $\langle \nabla F(x_t), e_t \rangle$, which is controlled via Young's inequality and Theorem 2. Because the softmax heavily weights clients whose directions align with the proxy (which in turn aligns with v_{true} by Assumption 5), misaligned drifts receive exponentially smaller weights, leading to $\Gamma_{\text{DAFA}} < \Gamma_{\text{FedAvg}}$. In a simple two-cluster model, we explicitly show $\Gamma_{\text{DAFA}} \leq (1 - \tau)\Gamma_{\text{FedAvg}}$ with $\tau = \Omega(\gamma\rho/N) > 0$.

7 Experiments

We empirically evaluate DAFA on three standard FL benchmarks to answer:

1. Does DAFA improve final performance and convergence speed under non-IID data?
2. Does empirical DSNR behavior agree with our theory?
3. How sensitive is DAFA to hyperparameters γ and β?
4. Are the key assumptions (proxy reliability, magnitude bounds) reasonable in practice?

7.1 Experimental Setup

Datasets & Models

- **CIFAR-10**: 50k training and 10k test images (32×32×3, 10 classes). Model: ResNet-18 with standard data augmentation.

- **FEMNIST**: Extended MNIST dataset with 62 classes, naturally partitioned by writer (client). Model: CNN with two convolutional layers (32, 64 filters, kernel 5) and two fully connected layers (512 units, 62-way output).
- **Shakespeare**: Federated language modeling dataset with 2.2M characters across 1,115 clients. Model: 2-layer LSTM with 256 hidden units and 200-dim embeddings.

Data Heterogeneity

- **CIFAR-10**: We simulate label-skewed non-IID partitions using a Dirichlet distribution $\mathrm{Dir}(\alpha)$ over class proportions with $\alpha \in \{0.5, 0.1\}$.
  - $\alpha = 0.5$: moderate non-IID.
  - $\alpha = 0.1$: severe non-IID (clients are nearly single-label).
- **FEMNIST & Shakespeare**: Use the official natural user partitions, which are inherently non-IID.

Training Protocol

- Clients: 100 (CIFAR-10), 200 (FEMNIST), 100 (Shakespeare).
- Client sampling: each round samples 10% clients uniformly without replacement.
- Local epochs per round: $K = 5$ (CIFAR-10), $K = 10$ (FEMNIST), $K = 2$ (Shakespeare).
- Batch size: 32.
- Client learning rate: 0.01 with cosine decay.
- Global step size $\eta_{\text{global}} = 1.0$.
- Communication rounds: $T = 200$.
- DAFA hyperparameters: default $\gamma = 1.0$, $\beta = 0.9$.
- Each experiment is repeated with 5 random seeds; we report mean $\pm$ standard deviation.

7.2 Baselines

We compare against:

1. **FedAvg** [1]: uniform averaging (size-weighted).
2. **FedProx** [2]: proximal term with $\lambda = 0.01$.
3. **SCAFFOLD** [3]: control variates, following open-source implementation.

4. **FedNova** [4]: normalized gradient aggregation.

5. **Directional Baseline (Dir-Weight)**: a naive cosine-similarity-based weighting without momentum proxy, for ablation.

6. FedAvgM [XX]: FedAvg with server-side momentum, to isolate the effect of our directional weighting from the benefits of standard server momentum.

7. FedAdam [XX]: Adaptive server-side optimization from the FedOpt framework.

All baselines share identical local optimizers and hyperparameters where applicable.

7.3 Main Results: Accuracy and Perplexity

Table 1 reports the final performance after 200 communication rounds.

Table 1: **Final test performance (mean ± std over 5 seeds).**
CIFAR-10 and FEMNIST: accuracy (%).
Shakespeare: perplexity (lower is better).

Dataset	Heterogeneity	FedAvg	FedProx	SCAFFOLD	FedNova	**DAFA (Ours)**
CIFAR-10	Dir($\alpha = 0.5$)	84.8 ± 0.2	86.1 ± 0.3	86.6 ± 0.3	85.6 ± 0.2	**88.2 ± 0.2**
CIFAR-10	Dir($\alpha = 0.1$)	77.7 ± 0.3	80.7 ± 0.3	81.7 ± 0.3	79.0 ± 0.3	**85.1 ± 0.3**
FEMNIST	Natural partition	90.4 ± 0.2	91.7 ± 0.2	92.3 ± 0.2	90.8 ± 0.2	**93.4 ± 0.2**
Shakespeare	Natural partition	15.1 ± 0.1	14.4 ± 0.1	14.6 ± 0.1	14.7 ± 0.1	**13.4 ± 0.1**

Observations

- Under **severe non-IID** (CIFAR-10, $\alpha = 0.1$), DAFA improves accuracy by **7.4%** over FedAvg and by **3.4%** over the best non-directional baseline (SCAFFOLD), demonstrating strong robustness to heterogeneity.
- On **FEMNIST**, DAFA improves over FedAvg by **3.0%** and over SCAFFOLD by **1.1%**, indicating that its benefits extend beyond synthetic Dirichlet partitions to real-world user partitions.
- On **Shakespeare**, DAFA reduces perplexity by **1.7** compared to FedAvg and by **1.2** compared to the best baseline, indicating more stable sequence modeling.

Convergence Curves.

Figure 1 (CIFAR-10, $\alpha = 0.1$) shows test accuracy vs. communication rounds. DAFA converges faster and exhibits fewer oscillations, reaching 80% accuracy in roughly 60 rounds vs. 90 rounds for SCAFFOLD and >110 rounds for FedAvg.

7.4 Empirical DSNR and Update Variance

To probe Theorem 1, we approximate the true global direction v_{true}^t using a full-batch gradient computed on a **small centralized validation subset** (2% of the full training data, not used for training). At round t, we compute:

- Empirical DSNR:

$$\widehat{DSNR}(\Delta_{agg}^t) = \frac{\langle \Delta_{agg}^t, \hat{v}_{true}^t \rangle^2}{\|\Delta_{agg}^t - \hat{\Delta}_*^t\|^2 + \varepsilon},$$

  with a small ε for numerical stability.

- Update variance across clients:

$$\widehat{\mathrm{Var}}(\Delta^t) = \frac{1}{|S_t|} \sum_{i \in S_t} \|\Delta_i^t - \Delta_{\mathrm{agg}}^t\|^2.$$

Figure 2 (CIFAR-10, $\alpha = 0.1$):

- **Left (DSNR)**: FedAvg's DSNR steadily degrades as training proceeds; SCAFFOLD and FedProx moderately mitigate the drop. DAFA maintains a DSNR curve consistently **1.5–2.0× higher** than FedAvg and **30–40%** higher than SCAFFOLD.
- **Right (Variance)**: DAFA's update variance is **37% lower** than FedAvg on average, directly supporting the variance reduction effect predicted by our analysis.

These results empirically validate that **directional alignment increases a DSNR-related metric**, and that DAFA is indeed suppressing destructive interference among client updates.

7.5 Ablation Study on Temperature γ

We study DAFA's sensitivity to the softmax temperature γ on CIFAR-10 with $\alpha = 0.1$. We vary $\gamma \in \{0, 0.5, 1.0, 2.0, 5.0\}$, where $\gamma = 0$ reduces DAFA to FedAvg.

Table 2: Effect of γ on CIFAR-10 ($\alpha = 0.1$).

γ	0.0 (FedAvg)	0.5	**1.0**	2.0	5.0
Accuracy (%)	77.7	82.1	**85.1**	83.5	80.2
Empirical DSNR	1.23	1.67	**1.98**	2.05	2.10

Findings

- Increasing γ from 0 to 1.0 monotonically increases both accuracy and DSNR, indicating effective suppression of misaligned updates.
- When γ becomes too large ($\geq$5.0), DSNR continues to increase slightly, but accuracy **drops sharply**, as aggregation degenerates toward selecting only a few clients (high bias regime).
- The optimal regime is **moderate temperatures** ($\gamma \in [0.5, 2.0]$), which aligns well with Theorem 2's prediction that bias grows linearly with γ.

This provides a **practical guideline**: choose γ in a moderate range to balance alignment and bias.

7.6 Ablation on Momentum Coefficient β

We further vary the proxy momentum $\beta \in \{0.0, 0.5, 0.9\}$ on CIFAR-10 ($\alpha = 0.1$).

- $\beta = 0.0$: no temporal smoothing, proxy equals current FedAvg direction.
- $\beta = 0.5$: moderate smoothing.
- $\beta = 0.9$: strong smoothing (default).

Results (omitting standard deviations for brevity):

β	Accuracy (%)	Empirical $\rho = E[\langle v_*, v_{true} \rangle]$
0.0	83.2	0.71
0.5	84.3	0.79
0.9	**85.1**	**0.83**

Larger β improves proxy reliability and slightly improves accuracy, but the gains plateau beyond 0.9. We thus recommend $\beta \in [0.8, 0.9]$ as a robust default across tasks.

7.7 Validation of Key Assumptions

We empirically validate Assumptions 3 and 5 on CIFAR-10 ($\alpha = 0.1$):

- **Proxy reliability (Assumption 5).** We estimate

$$\rho_t = \langle v_*^t, \hat{v}_{true}^t \rangle$$

using the same centralized validation gradient as in Section 7.4. Over 200 rounds, we observe ρ_t stays above 0.8 for 85% of rounds, with mean 0.83 ± 0.05, confirming the assumption holds in standard training regimes.

- **Update magnitude lower bound (Assumption 3).** We monitor $\min_{i \in S_t} \|\Delta_i^t\|$ across rounds. For most of training, it remains above 0.04; near convergence, a small fraction of rounds violate $\mu = 0.01$, but we apply a practical safeguard: when $\|\Delta_i^t\| < \mu$, we **skip normalization and treat** $s_i^t = 0$, effectively not over-amplifying noisy directions. With this implementation detail, Assumption 3 holds for the normalized directions used in DAFA.

These results suggest that our theoretical assumptions are **realistic** under typical, non-pathological FL scenarios.

7.8 Comparison to a Naive Directional Baseline

To disentangle the effect of using a **momentum proxy** vs. raw per-round FedAvg, we include a naive **Dir-Weight** baseline:

- Dir-Weight: same as DAFA but with $v_*^t = \Delta_{\text{FedAvg}}^t / \|\Delta_{\text{FedAvg}}^t\|$ (i.e., $\beta = 0$) and fixed $\gamma = 1.0$.

On CIFAR-10 ($\alpha = 0.1$):

- Dir-Weight achieves 83.2% accuracy (vs. 85.1% for DAFA).
- Empirical proxy reliability drops to $\rho \approx 0.71$ (vs. 0.83).

This indicates that **temporal smoothing** plays a crucial role in stabilizing the proxy direction and fully realizing the benefits of direction-aware aggregation.

8 Discussion and Limitations

Directional Viewpoint. DAFA demonstrates that explicitly accounting for directional reliability of client updates can significantly improve optimization behavior under heterogeneity. Our DSNR-based analysis provides a principled lens for understanding why naive uniform averaging fails in highly non-IID regimes.

Practicality. DAFA is easy to deploy in real FL systems:

- It is **purely server-side**, requiring no change to client code.
- It introduces no extra communication rounds or client state.
- Computational overhead is negligible relative to model training.

Hyperparameter Robustness. Our experiments show that:

- A moderate γ (e.g., 0.5–2.0) and high β (e.g., 0.8–0.9) work robustly across benchmarks.

- DSNR curves provide an intuitive diagnostic tool: if DSNR is low and volatile, increasing γ may help; if bias dominates (accuracy drops despite high DSNR), reducing γ is advisable.

Limitations.

1. **Pathological Heterogeneity & Adversaries.** When the majority of clients are adversarial or extremely misaligned, FedAvg's momentum may no longer be a reliable proxy. In such cases, DAFA could amplify malicious directions. Integrating DAFA with robust aggregation (e.g., median or clipped mean) and robust proxy estimation is an important future direction.

2. **Near-Convergence Regime.** As training approaches a stationary point, update magnitudes shrink and directional estimation becomes noisy. Our practical safeguard (thresholding small updates) alleviates this, but a more refined analysis for the late training phase is left to future work.

3. **DSNR Measurement and Privacy.** In this work, we compute empirical DSNR using a small centralized validation set to theoretically validate our surrogate bound. However, requiring centralized data violates strict FL privacy constraints. A promising future direction is to develop **decentralized DSNR proxies**—for instance, by measuring alignment across historical aggregated updates or utilizing temporal gradient differences—allowing dynamic hyperparameter tuning (γ, β) without centralized data access.

9 Conclusion

We identified directional dispersion as a key bottleneck for federated learning under data heterogeneity. We proposed **DAFA**, a simple yet principled aggregation rule that leverages **directional alignment** with a momentum-based proxy to adaptively reweight client updates.

On the theoretical side, we introduced **Directional Signal-to-Noise Ratio (DSNR)** and derived a **surrogate lower bound** that DAFA explicitly optimizes. We proved that DAFA controls aggregation bias, improves a heterogeneity-related constant in non-convex convergence bounds, and recovers FedAvg as a special case.

On the empirical side, extensive experiments on CIFAR-10, FEMNIST, and Shakespeare demonstrate that DAFA significantly improves convergence stability, final accuracy, and empirical DSNR under severe non-IID settings, with negligible overhead and no extra communication.

We believe this work opens a promising research direction on **direction-aware federated optimization**, and that incorporating directional reliability can further benefit robust FL, personalized FL, and federated multi-task learning.